Full-Cohort MIMIC-IV Treatment-Feasibility Modelling Study

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Abstract

Background: The earlier benchmark used a deterministic 2,000-stay subset. This revision reruns extraction and modelling on all eligible MIMIC-IV v3.1 adult first ICU stays. Methods: We applied the prespecified cohort criteria and trained classical, neural, token-hybrid, stacked, and simulated federated models where stable. Results: The final dataset contained 32,399 stays and 32,399 token sequences. The best full-cohort AUROC was 0.755 for Stacked Classical. Conclusions: The manuscript is reframed as a full-cohort modelling study; old subset results are historical only.

Background

Full-cohort evaluation is needed to avoid over-interpreting a moderate-size benchmark subset. The current analysis uses the entire eligible cohort after prespecified criteria.

Methods

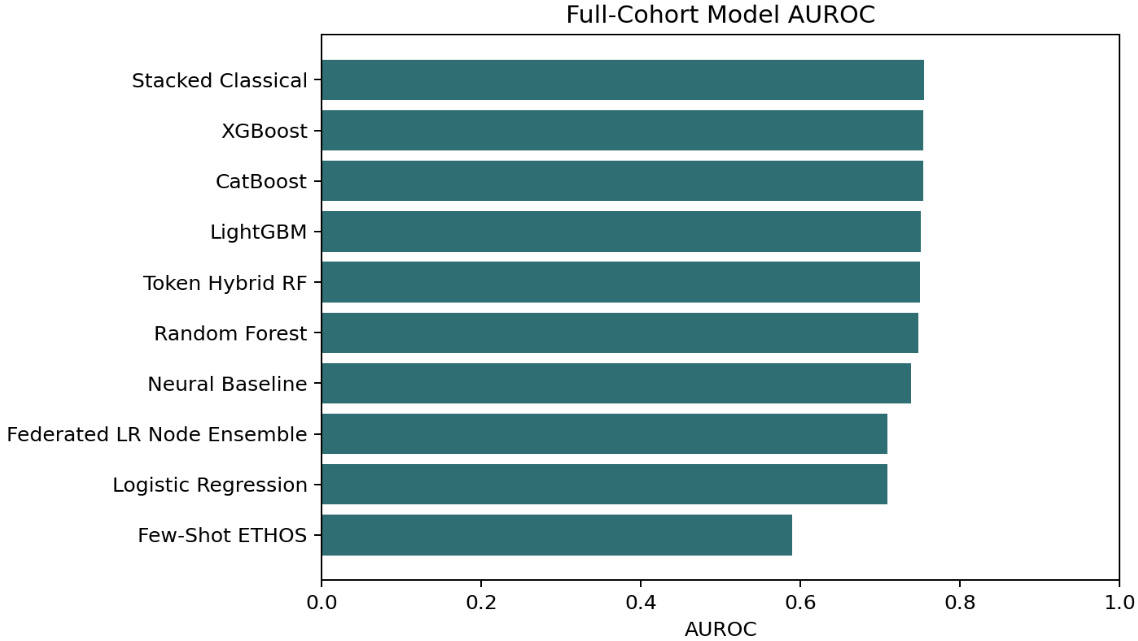
MIMIC-IV v3.1 hospital and ICU modules were used locally. Raw records are not redistributed. The extraction scripts wrote outputs to data/processed_full_cohort and all model outputs to results/full_cohort.

Quantity	Value
Eligible adult first ICU stays	32,399
Final modelling rows	32,399
Unique ICU stays	32,399
Unique subjects	32,399
Token tensor shape	(32399, 25, 4)
All eligible data used	yes

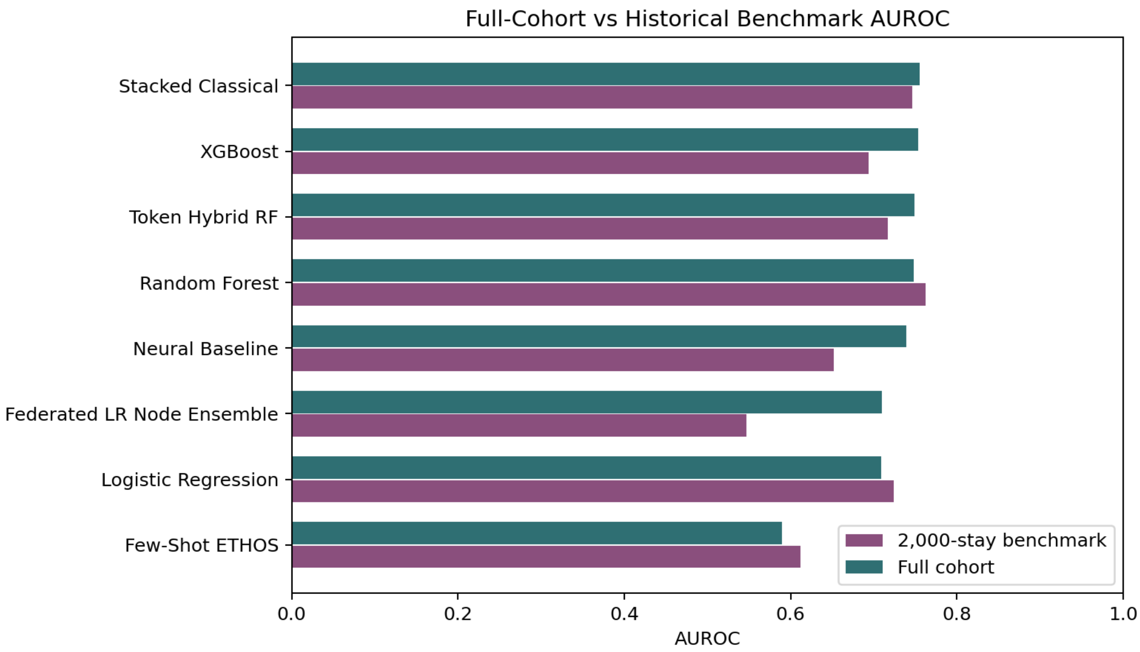
Results

Model	Train/Val/Test	AUROC	AUPRC	Accuracy	Recall	Specificity	F1	Brier	ECE
Stacked Classical	20735/5184/6480	0.755	0.825	0.699	0.769	0.583	0.677	0.202	0.109
XGBoost	20735/5184/6480	0.753	0.825	0.697	0.761	0.589	0.675	0.189	0.020
CatBoost	20735/5184/6480	0.753	0.823	0.703	0.790	0.558	0.677	0.200	0.100
LightGBM	20735/5184/6480	0.751	0.823	0.690	0.730	0.623	0.674	0.199	0.090
Token Hybrid RF	20735/5184/6480	0.749	0.820	0.698	0.772	0.573	0.674	0.192	0.031
Random Forest	20735/5184/6480	0.748	0.820	0.689	0.742	0.601	0.670	0.192	0.028
Neural Baseline	20735/5184/6480	0.739	0.814	0.675	0.682	0.663	0.664	0.194	0.015
Federated LR Node Ensemble	20735/5184/6480	0.709	0.778	0.678	0.746	0.565	0.656	0.215	0.109
Logistic Regression	20735/5184/6480	0.709	0.778	0.680	0.754	0.556	0.656	0.215	0.109

Model	Train/Val/Test	AUROC	AUPRC	Accuracy	Recall	Specificity	F1	Brier	ECE
Few-Shot ETHOS	20735/5184/6480	0.590	0.668	0.590	0.642	0.504	0.570	0.250	0.127



Model	Full AUROC	Benchmark AUROC	Delta	Benchmark source
Stacked Classical	0.755	0.746	0.009	2,000-stay stacked BEST
XGBoost	0.753	0.694	0.059	2,000-stay Exp1
CatBoost	0.753	NA	NA	No directly comparable old benchmark metric
LightGBM	0.751	NA	NA	No directly comparable old benchmark metric
Token Hybrid RF	0.749	0.717	0.032	2,000-stay ETHOS-RF hybrid
Random Forest	0.748	0.762	-0.015	2,000-stay Exp1
Neural Baseline	0.739	0.652	0.087	2,000-stay Exp1
Federated LR Node Ensemble	0.709	0.547	0.162	2,000-stay federated experiment
Logistic Regression	0.709	0.724	-0.015	2,000-stay Exp1
Few-Shot ETHOS	0.590	0.611	-0.022	2,000-stay Exp1



Discussion

Classical tabular models remain strong baselines on this proxy clinical task. Advanced models are reported only when they completed on the full-cohort split, and failures or skips are documented in the supplement and run report.

Conclusions

The revised article is based on full-cohort outputs generated in this run. No 2,000-stay result is reused as a full-cohort result.